Building an AI Voice Agent to Transform Customer Support

In the fast-growing D2C and e-commerce sector, delivering fast, consistent, and empathetic customer support at scale is one of the most pressing operational challenges.

ShopNow, a fictional mid-size D2C brand processing 40,000+ orders a month across India, currently relies on 35 human agents available 9 AM to 9 PM. Average wait time is 8 minutes, first-contact resolution is 52%, and CSAT sits at 3.1/5.

The majority of incoming calls are repetitive Tier-1 queries -- order status, return requests, payment issues, and delivery complaints -- that do not require human judgment but consume the bulk of agent time. Meanwhile, customers calling after 9 PM, or in languages other than English, receive no support at all.

ShopNow's leadership has greenlit an AI-first support transformation. You are building Version 1.0.

What does the Voice Agent need to do?

The voice agent must handle the full lifecycle of a customer support call -- from greeting to resolution or escalation -- in real time, across languages, and with genuine conversational intelligence. It is not a phone tree. It is not a keyword-matching FAQ bot. It is a system that listens, understands context across multiple turns, detects when a customer is frustrated, and knows when to step aside for a human.

How do we handle customer support today? What are the challenges, if any?

Once a customer calls in, the current process involves two main steps.

Step 1 - Issue Identification: The agent pulls up the customer's order history, identifies the issue type, and manually looks up relevant data (shipping status, return eligibility, payment records). This is time-consuming and error-prone.

Step 2 - Resolution or Escalation: The agent attempts to resolve the issue on the call. If the issue is complex or the customer is distressed, it is escalated to a senior agent or specialist. This handoff is often poorly contextualised -- the new agent starts from scratch.

These processes are slow, inconsistent, and unavailable outside business hours. Multilingual customers are particularly underserved. The goal is to automate Tier-1 resolution entirely, and to make escalations fast, context-rich, and empathetic.

What is the solution we are looking to create?

We are looking for an AI Voice Agent solution that acts as a digital customer support representative, delivering the following:

- The agent must handle at minimum five distinct customer intents: **order status, return and refund initiation, payment issue reporting, delivery complaint, and product query.** Each must support multi-turn dialogue -- the agent must remember what was said earlier in the same call.
- The agent must support at least three languages -- **Hindi, English, and one regional language of your choice.** It must automatically detect the customer's language without asking them to press 1 for Hindi, and handle code-switching mid-conversation.
- The agent must perform real-time sentiment detection at the utterance level and apply intelligent escalation logic. Define and defend your escalation thresholds. When the handoff occurs, the human agent receiving the call must get an instant brief: customer name, issue summary, sentiment history, and recommended tone.
- At the end of each day, the system must generate a **detailed operations report** for the support leadership team: call volume by intent, resolution rates, escalation breakdown, sentiment trends by language and time of day, and a list of the top unresolved query patterns for the next model iteration.

Technical Environment (only suggestions, you can use other solutions as well):

- **Speech-to-Text / Text-to-Speech:** Sarvam AI, Bhashini, Google Cloud STT/TTS, Azure Speech, or equivalent
- **NLU / Conversation Engine:** Rasa, Dialogflow, LangChain, OpenAI function calling, or custom-built
- **Backend:** Python (FastAPI, Flask, or Django)
- **Frontend / Dashboard:** React.js, Streamlit, Power BI, or equivalent
- **Infrastructure:** Docker for packaging; cloud deployment optional but encouraged
- **Must work on:** Windows and macOS environments

Provided Data:

Participants will receive the following synthetic datasets to build and test the system:

1. Synthetic Call Transcript Dataset -- 500 multi-turn transcripts across 8 intent categories in Hindi and English, with labelled intents, entities, and sentiment tags
2. Order and Returns Schema -- Simulated backend database your agent can query for order status, return eligibility, and refund status
3. Escalation Log Dataset -- Historical escalation records with reason codes, resolution outcomes, and handle times
4. Support Performance Baseline -- Pre-AI metrics (CSAT, FCR, AHT) to benchmark your system's projected improvement against

Expected Deliverables:

1. **A working AI Voice Agent.** Please share a 10-slide PPT and a video of the agent handling at least three complete call scenarios, including one multilingual call and one escalation.

2. **Report and Documentation** explaining: System architecture | Full code with setup instructions | Multilingual approach | Sentiment and escalation logic | Sample daily operations report

3. **A live demonstration** for selected participants, showing the agent handling real-time calls from judges -- including edge cases and deliberate attempts to break the system.

Evaluation Criteria:

Criteria	Weight	Description
Technical Quality	40%	Accuracy of speech recognition, intent understanding, and conversation processing
Performance & Reporting	10%	Ability to handle concurrent calls and generate meaningful daily reports
User Experience	20%	Clear, empathetic, and natural interactions across languages
Innovation	10%	Creative solutions to improve customer resolution and engagement
Documentation & Presentation	10%	Clear explanation, easy-to-follow instructions, and video of the bot in action
Code Quality and Code Documentation	10%	Well structured, readable code with thorough inline comments and architecture documentation